User Guide: Library Space Management Tool

Distribution Analysis & Navigation Algorithm (D.A.N.A.) Framework

Project Creator:	Dana Makinen	**Supported Metrics:**	Imperial (Inches) / Metric (Centimeters)
Component:	Library Space Management Tool	**Input Data Source:**	Exported Range Summary CSV

Welcome to the Library Space Management Tool (LSMT). This document is a simple guide for first-time users & non-technical staff, custom-built to help library staff analyze current and projected shelf space utilization across different collections. The tool automatically figures out exactly how much shelf space to shift from over-allocated (surplus) ranges to crowded (needy) ranges while minimizing physical book movement.

1. First-Time Setup (Prerequisites)

Because this application executes locally on your own computer rather than running on an external website, it requires a free software component named Python. If your computer does not have Python installed yet, please complete this single prerequisite step first:

CRITICAL INSTALLATION STEP

1. Go to your web browser and visit: `python.org/downloads`
2. Click the large designated download button to download Python install manager.
3. Open the downloaded installation file.
4. Permit Python to install on your system.
5. A command prompt window will open and you will be asked four times to answer [y,n]. You must answer **"y"** for the first three questions.

2. Organizing and Launching the Tool

To run the program seamlessly, you will need several files that are found as part of the D.A.N.A. framework download folder along with the Data Collection Workbook. Ensure the following files are stored together in the exact same folder on your computer (for example, in a new folder on your Desktop named "Library Tool"):

- `launch_library_tool.bat` (The automated setup & launch script)
- `library_app.py` (The core application code)
- `Example CSV Data Collection Workbook - 11th Floor 2026 - Range Summary.csv` (A sample dataset provided for practicing and playing around with the tool's features)

How to Launch:

1. Double-click the file named `launch_library_tool.bat`.
2. A black text window (Command Prompt) will open up. Press any key on your keyboard to continue when prompted.
3. **First Run Only:** The script will automatically connect to the internet and install the required data analysis frameworks (Streamlit, Pandas, Plotly). This takes about 30-60 seconds. Subsequent launches will take only a couple seconds.
4. Once finished, your web browser will automatically open a new tab containing the tool's interactive dashboard at `http://localhost:8501`.

> **HOW TO CLOSE THE TOOL**
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> When you are done using the application, simply close your web browser tab and close the black text window on your desktop. Closing that window safely turns off the application background process.

3. Step-by-Step Dashboard Guide

The web page interface is split into clear, ordered sections. Follow them from top to bottom:

Configuration:

- **Select System of Measurement:** Click either Imperial (Inches) or Metric (Centimeters). The tool instantly adjusts all mathematical formulas, tables, and charts to reflect your selection.
- **Upload Range Summary CSV File:** Click the upload area to select your library's shelf data file (or use the provided `Example CSV Data Collection Workbook - 11th Floor 2026 - Range Summary.csv` to test the dashboard features). This file should contain columns detailing your ranges (e.g., 1a, 1b, 2a), current total shelf capacities, and currently occupied or available space lengths.
- **Understanding Collection Configurations:** The application displays two editable text boxes formatted in text blocks called JSON. These control your targets. Do not delete the quotation marks or curly brackets; simply change the numerical values to match your specific library targets.
- **Collection Target Empty Percentages:** Defines how much baseline "breathing room" or buffer space each collection should retain (e.g., 0.25 means a target of 25% empty space).
- **Collection Annual Growth Rates:** Input how many inches or centimeters of physical books are added to each side of a range per year to map out growth forecasts.
- **Number of Years to Project:** Adjust this number (e.g., 5 years) to simulate how your shelves will hold up over time based on your annual growth rates.

VARYING SHELF LENGTHS

All range sides will need an average shelf length entered. This is important so that varying shelf sizes in a range will be properly accounted for. If a range side contains shelves of differing physical lengths, you must enter the average shelf length for that specific range row under the "Average Shelf Length" column in your Data Collection workbook before downloading the CSV file.

Example: If a range side has 15 shelves at 35.5” and 2 shelves at 24” the Total Capacity is 580.5”. To find your average value for that row, compute 580.5/17 which gives an average of 34.15”.

4. Reviewing and Extracting Results

Analysis Results:

As soon as a valid file is loaded, the application instantly outputs data dashboards summarizing two distinct scenarios:

Metric Item	Current Year Dashboard	Projected Year Dashboard
What it Analyzes	Calculates the immediate shift maneuvers required to satisfy your target empty space parameters today.	Forecasts shelves after compounding your annual book growth rates over your specified timeline.
Proposed for Shift	The volume of adjustments required in the future to keep ahead of structural overcrowding.	
Remaining Unfulfilled Need	The total volume of shelf length recommended to be packed and physically moved to alternative ranges. If greater than 0, it gives a clear warning indicating that your library is physically out of room to achieve target space.	Highlights when and where future book additions will completely overwhelm a collection's footprint.

GREEN & ORANGE NOTIFICATIONS

The program evaluates the scenario automatically. A Green Banner means your available shelf footprint completely handles the parameters. An Orange Banner acts as an alert that there is a spatial deficit, listing the unfulfilled shelf space length remaining.

Detailed Shift Impact Report (Targeted Range Analysis):

In order to see precise instructions for a set group of ranges, use the text boxes under Section 3 to filter your view:

1. Type a starting position into the **Start Range ID** box (e.g., 1a) and an ending range into the **End Range ID** box (e.g., 10b).
2. Click the **Run Localized Shift Analysis** button.
3. The system filters out the rest of the library and produces an actionable table showing exactly how much space is planned to move Out of or In to each individual range side, alongside its initial space versus its target layout.
4. Click **Reset to Full Library Analysis** at any time to return to the library-wide views.
5. **Download the Report:** To extract your data, click the **Download Shift Impact Report (CSV)** button located right below the data table. This generates a local spreadsheet file tailored precisely to your filtered ranges.

> **FIRST-TIME USER RECOMMENDATION**
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> For first-time users, it is highly recommended to try a targeted shift spanning only a couple of ranges to gain familiarity working with the program and to understand how physical real-world environments impact final results.

> **OPERATIONAL REALITY CHECK**
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> Although this tool creates highly accurate predictive shifting guides, it cannot account for unanticipated physical variations between mathematical estimates and real-world shelf transitions. As a professional rule of thumb, it is recommended to add an extra 5% of buffer space on top of the LSMT's final estimates when executing physical moves.

Visual Charts:

- **Collection Performance Bar Chart:** Visually displays which exact collections are running out of space (bars pointing Up indicate space needed; bars pointing Down indicate extra surplus capacity).
- **Distribution of Range Fullness Histograms:** Provides a side-by-side graphical look at your shelf distribution statuses across three states: Initial layout, Post-Shift layout (if you complete the recommended moves), and Projected future state.